NAME:PAREENITA A.SHIRSATH BE AI&DS ROLL.NO: 49

AIFBA EXPERIMENT : 05

```
import pandas as pd
import numpy as np
data = {'Date': pd.to_datetime(['2024-01-01', '2024-01-02', '2024-01-03',
                                '2024-01-04', '2024-01-05']),
    'Price': [10, 12, 15, 14, 16]
}
df = pd.DataFrame(data).set_index('Date')
short_window = 2
long_window = 3
df['SMA_Short'] = df['Price'].rolling(window=short_window).mean()
df['SMA_Long'] = df['Price'].rolling(window=long_window).mean()
df['Signal'] = 0.0
df.loc[df.index[short_window:], 'Signal'] = np.where(
    df['SMA_Short'][short_window:] > df['SMA_Long'][short_window:],
    1.0,
    0.0)
df['Positions'] = df['Signal'].diff()
initial_capital = 10000
portfolio = initial_capital
shares = 0
for i in range(len(df)):
    if df['Positions'].iloc[i] == 1:
        shares = portfolio // df['Price'].iloc[i]
        portfolio -= shares * df['Price'].iloc[i]
        print(f"Buy {shares} shares at {df['Price'].iloc[i]} on {df.index[i]}")
    elif df['Positions'].iloc[i] == -1:
        portfolio += shares * df['Price'].iloc[i]
        print(f"Sell {shares} shares at {df['Price'].iloc[i]} on {df.index[i]}")
        shares = 0
portfolio += shares * df['Price'].iloc[-1] # This line was likely intended to account for any remaining shares at the end
print("\nFinal Portfolio Value:", portfolio)
print("Profit/Loss:", portfolio - initial_capital)
print("\nDataFrame with Signals:")
print(df)
```

```
Buy 666 shares at 15 on 2024-01-03 00:00:00
Sell 666 shares at 16 on 2024-01-05 00:00:00

Final Portfolio Value: 10666
Profit/Loss: 666

DataFrame with Signals:
            Price  SMA_Short   SMA_Long  Signal  Positions
Date
2024-01-01     10        NaN        NaN     0.0        NaN
2024-01-02     12       11.0        NaN     0.0        0.0
2024-01-03     15       13.5  12.333333     1.0        1.0
2024-01-04     14       14.5  13.666667     1.0        0.0
2024-01-05     16       15.0  15.000000     0.0       -1.0
```